below such a number. How do you know where the crossover point is? This is a problem. My colleagues and I worked with around 20 million pieces of financial data. We all had the same data set, yet we never agreed on exactly what the exponent was in our sets. We knew the data revealed a fractal power law, but we learned that one could not produce a precise number. But what we did know—*that the distribution is scalable and fractal*—was sufficient for us to operate and make decisions.

The Problem of the Upper Bound

Some people have researched and accepted the fractal "up to a point." They argue that wealth, book sales, and market returns all have a certain level when things stop being fractal. "Truncation" is what they propose. I agree that there is a level where fractality *might* stop, but where? Saying that there is an upper limit *but I don't know how high it is,* and saying *there is no limit* carry the same consequences in practice. Proposing an upper limit is highly unsafe. You may say, Let us cap wealth at $150 billion in our analyses. Then someone else might say, Why not $151 billion? Or why not $152 billion? We might as well consider that the variable is unlimited.

Beware the Precision

I have learned a few tricks from experience: whichever exponent I try to measure will be likely to be overestimated (recall that a higher exponent implies a smaller role for large deviations)—what you see is likely to be less Black Swannish than what you do not see. I call this the masquerade problem.

Let's say I generate a process that has an exponent of 1.7. You do not see what is inside the engine, only the data coming out. If I ask you what the exponent is, odds are that you will compute something like 2.4. You would do so even if you had a million data points. The reason is that it takes a long time for some fractal processes to reveal their properties, and you underestimate the severity of the shock.

Sometimes a fractal can make you believe that it is Gaussian, particularly when the cutpoint starts at a high number. With fractal distributions, extreme deviations of that kind are rare enough to smoke you: you don't recognize the distribution as fractal.

The Water Puddle Revisited

As you have seen, we have trouble knowing the parameters of whichever model we assume runs the world. So with Extremistan, the problem of